



Quarterly Operations Review

Supply Chain Optimization & Asset Health

Transitioning to Predictive Maintenance and Dynamic Distribution

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Executive Summary



The Objective

Transition KPC's distribution nodes from reactive, fixed-interval maintenance and static inventory models to an integrated, predictive supply chain framework.



The Methodology

Integrated machinery health monitoring with demand forecasting and linear distribution optimization.



The Impact

The proposed model achieves >98% anomaly classification accuracy, guarantees a 95% service availability across the network, and eliminates emergency freight expenses during peak demand.

Safety & Equipment Health



Predictive Baseline

Transitioned from time-based maintenance to condition-based monitoring.



Telemetry Integration

Continuous evaluation of vibration (g), temperature (°C), and motor current against Flowgard's physics-referenced pressure residuals.



Model Performance

Successfully achieved an **84.3%** failure-detection sensitivity and a low **2.4%** false alarm rate in testing.

Proactive Risk Mitigation



Early Intervention

Regression models estimate Remaining Useful Life (RUL) with a Mean Absolute Error of under 105.9 hours (approx. 4.4 days).



Safety Compliance

Proactive mitigation directly prevents severe pressure excursions and mechanical anomalies (e.g., seal/bearing degradation).



Operational Continuity

Shifting failures from unplanned to scheduled eliminates the reliance on technician overtime and expedited logistics.

Supply Chain Forecasting



Demand Modeling

Deployed a Prophet time-series forecasting model to project daily product throughput for a 60-day horizon.



External Regressors

Successfully integrated holiday and seasonal travel peaks (e.g., April and December) to account for demand spikes.



Validation

Baseline models establish clear boundaries for expected volume vs. peak season throughput.

Inventory Optimization: Safety Stock & ROP



Data-Driven Buffers

Calculated Safety Stock (SS) using a 95% service level ($Z = 1.645$) and a 5-day lead time.



Dynamic Reorder Points (ROP)

Replaced static inventory thresholds with forecast-driven ROPs to align with dynamic demand patterns.



Simulation Results

Stress-testing proves operating *without* this buffer causes recurring stockouts; utilizing it guarantees 95% availability.

Distribution Network Optimization



The Problem

Supplying 3 regional depots (Nakuru, Eldoret, Kisumu) from 2 primary terminals (Mombasa, Nairobi) efficiently.



PuLP Implementation

Formulated a cost-minimization algorithm respecting maximum supply capacities and forecasted depot demands.



The Output

Generates an optimal weekly shipping plan that allocates exact fuel volumes to specific routes, systematically minimizing total transport costs per litre.

Strategic Recommendations



Automate Inventory Alerts

Integrate ROP triggers directly into the ERP.

Estimated ROI: 15% reduction in holding overhead.



Dynamic Allocation

Adopt PuLP routing algorithms for all weekly tanker and pipeline scheduling.

Estimated ROI: KES 4.2M in annual logistics savings.



Synchronize Maintenance

Tie safety stock levels to pump asset health scores to buffer expected downtimes.

Estimated ROI: Elimination of unplanned throughput stoppages.